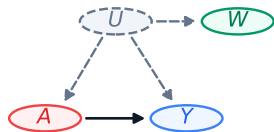
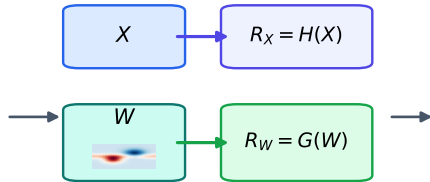


1. Single proxy structure



W : pre-treatment proxy
 A does not cause W

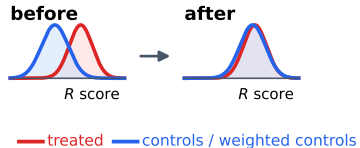
2. Representation



Required: R_W retains the
hidden-severity signal in W .

3. Proxy balancing

Weight controls to match treated
on $R = (R_X, R_W)$



If R_W is informative,
balancing R reduces imbalance in U .

Single Proxy Balancing: a pre-treatment proxy W supplies a representation that makes hidden-severity imbalance visible enough to balance.